Crucis: An Independent Framework for Auditing Causal-Discovery Systems

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Abstract

Final benchmark scores alone do not establish where a causal-discovery system's answer comes from: the numerical data, the variable names, the file layout, or the system's own rules. We introduce Crucis, an independent auditing framework with an open protocol for examining externally developed causal-discovery systems as black boxes or, where supported instrumentation is available, stage by stage. Its recorded applications hold numerical data fixed under name or layout controls, preserve available stage graphs, and report typed findings and explicit observation gaps within a stated regime and against a reference of stated reliability. Recorded audits illustrate distinct vulnerabilities in benchmarks, agents and individual pipeline stages, with each comparison retaining its own cohort and scoring rule. Benchmark scores reward layout: on CausalArena, an undecided-pair rule that follows the released column order adds 0.221 directed F1 to FCI. Name controls reveal reliance on semantics: swapping the names costs 3 of 4 entered systems at least 0.10 directed F1 under the declared, heterogeneous measurements; Causal-Copilot's comparison is exploratory. Its refinement lowers adjacency F1 from 0.584 to 0.321; paired stage diagnostics locate the main loss in bootstrap rather than model review. Every one of the 8 runs that reached its cycle step crashed. Public repositories separate the reusable framework, the research evidence and the companion reproduction.

Framework: https://github.com/crucis-lab/crucis
Research: https://github.com/uzairgheewala/cc-research

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1 Introduction

Many decisions turn on a causal question: if we change this, what else moves? When experiments are impossible or expensive, the structure is learned from observational data, and every discovery algorithm brings assumptions that an expert must check against the problem (Spirtes, Glymour and Scheines 2000). Language-model agents now promise to automate that expert. Causal-Copilot inspects the data, chooses and configures an algorithm, has a language model review the graph, and writes the report (Wang et al. 2025); several other systems let a model supply constraints, a causal order or edits to a data-driven graph (Khatibi et al. 2024; Shen et al. 2025; Vashishtha et al. 2025).

How far can such a system's claims be trusted? A benchmark score alone cannot establish where the answer came from, and there are known ways to earn one without the data: simulated benchmarks leak the causal order through the scale of the variables (Reisach, Seiler and Weichwald 2021), the column order of the input file changes what an algorithm learns (Kitson and Constantinou 2024), and language models reconstruct well-known graphs from the dataset and variable names alone (Srivastava et al. 2025). A recent position paper asks for audit trails that keep data, assumptions and agent recommendations apart (Zheng et al. 2026). Our goal is a reusable, independent way to audit externally supplied systems under explicit controls and to establish which comparisons their evidence supports.

Crucis combines system-connection and experiment-definition contracts with an auditing protocol. Its target is a continuous-integration check for causal claims: point it at a system through a small adapter and receive an audit report that says what the system gets right, where its answers come from, how its errors behave under controlled change, and whether its confidence means anything. The protocol has levels, each usable alone. L0 scores the final graph against a known one. L1 runs metamorphic controls: the same numbers under real, neutral and swapped names, with the columns shuffled and the file paths made opaque. These controlled comparisons test sensitivity to the information channel within the observed conditions. L2 traces a system stage by stage where its code is open, recording which step adds, deletes or reverses each edge and which model call caused it. L3 (claim provenance) and L4 (adversarial search) are designed and come later. Findings are typed from a closed list (name reliance, layout reliance, deletion of links the data support, crash, cost outlier, and so on), with their controls, available uncertainty estimates and decision rules. Confirmatory and exploratory findings are distinguished by the timing and scope of their pre-registration. A score is compared only within its regime (data only, or with interventions, or with declared knowledge), and a reference graph carries a reliability tier: exact when the data were simulated from it, contested when published versions disagree, provenance-only when nothing can be compared against it, and mismatched when it differs from the version a benchmark uses.

The recorded audit collection covers 5 language-model systems and the CausalArena and CausalBench benchmarks, with Causal-Copilot as a worked case traced stage by stage. The following findings have different measurement scopes and do not constitute a matched-cohort ranking.

- **Benchmarks reward shortcuts.** CausalArena's released files list the variables in causal order, and its scoring forces a direction on every undecided pair, broken by column index; FCI gains 0.221 directed F1 on the semantic family from that rule alone, and the published PC row (0.333) rises to 0.463 with an order-free collider rule on shuffled columns. On CausalBench, the main metric rewards ranking total effects: MeanDifference reaches 0.90–0.95 of the ceiling set by the test split's own knockdown effects, and the second metric, the false omission rate, does not separate it from random networks.
- **Names can influence agents.** With the numbers fixed, swapping the names costs 3 of the 4 systems that entered the pre-registered test (at least 5 paired scenarios each) at least 0.10 directed F1 (causaliq-knowledge 0.251, Causal-Copilot 0.167, MATMCD 0.122). Causal-Copilot measures its graph handed to cycle removal under committed-mark scoring on its earlier semantic suite; the other agents use final outputs and expectation scoring on a different semantic cohort. These cohorts share no scenarios. Causal-Copilot's runs predate the name-test pre-registration and are exploratory. The pipeline of Vashishtha et al. (2025) shows the smallest reported drop (0.020) within its tested conditions. Inside Causal-Copilot, BND-08's recorded label-following calculation counts distinct (scenario, directed-edge) pairs retained in completed swapped-name outputs, changed from bootstrap and matching their post-review pair state at final output, and adjacent in exactly one reference reading. Among these 101 informative pairs, 81 are adjacent under the label reading and 20 under the data reading. This classification tests adjacency, not agreement of edge direction; edges adjacent in both references or neither are outside this denominator. The decision rule comes from BND-08's pre-run specification, not a later finding threshold. However, that specification describes a pre-cycle endpoint while the recorded calculation consumes completed final outputs. We retain the counts as a descriptive diagnostic, without claiming confirmation of the specified endpoint.
- **An agent's own steps can undo its data.** MATMCD's final graph has lower committed-claim precision than its own data-only graph on the point estimate (0.6855 against 0.7199); the interval for the difference includes zero. The pre-registered point rule does not establish an interval excluding zero. Causal-Copilot's refinement lowers adjacency F1 from 0.584 (all 30 tables) to 0.321 (the 22 runs that finished). Paired completed-run diagnostics attribute the main loss to bootstrap within refinement, which makes no model call, rather than the language-model review, and all 8 runs that reached its cycle step crashed there (a missing key in the model's reply, 8 of 8).

The framework also sets a validation target: a system's claims should come with their source and with a confidence that survives these controls.

Contributions. (i) An open audit protocol with typed findings, regimes, truth tiers, pre-registered thresholds and reproducible certificates for supported recorded audits. (ii) A collection of scoped audits: name reliance across 5 systems, and layout and metric shortcuts in CausalArena and CausalBench. (iii) A stage-level account of Causal-Copilot from recorded runs, with every model call and intermediate graph kept. (iv) The code, the recorded runs and the source mappings behind every number in this report. The system-

connection and experiment-definition subsystems are implemented and tested; execution and collection remain in design. The broader benchmark and validation of the auditor itself remain development obligations. A companion project uses the audit's findings to rebuild Causal-Copilot's review step; it is audited through the same interface and is not reported here.

These contributions address linked evidentiary problems. Faithful connections preserve what each system actually received and returned; comparison contracts determine which controls and scores can support a finding; stage records can then localise a failure where observation permits. Without faithful connections the comparison can change a system's meaning, without controls a trace cannot isolate an information-channel effect, and without validation of the auditor an apparent failure may be a measurement defect. The recorded audits motivate and exercise parts of this combination. They do not by themselves establish general native interoperability or the auditor's scientific validity; the broader study must test both, across explicitly supported integrations.

This checkpoint covers the introduction, the prior work and the data. The protocol's full specification, the audit package and the results tables belong to the final report.

2 Prior work

2.1 How causal discovery is evaluated, and what benchmarks reward

Constraint-based algorithms such as PC and FCI recover causal structure from conditional-independence tests, and only up to an equivalence class under assumptions such as faithfulness (Spirtes, Glymour and Scheines 2000). Score-based search such as GES works over equivalence classes and, with a consistent score and a distribution that has a DAG perfect map, identifies the structure in the large-sample limit (Chickering 2002). Benchmarks with known graphs, on which these methods are compared, have been shown to carry shortcuts (Reisach, Seiler and Weichwald 2021; Kitson and Constantinou 2024). In common simulated settings the variance of the variables grows along the causal order, so sorting by variance matches continuous-optimisation methods (Reisach, Seiler and Weichwald 2021), and a scale-invariant version of the same artefact is also high in common settings and can be exploited (Reisach et al. 2023). The order of the columns in the input data alone changes the graph that hill-climbing search learns, typically more than a change in sample size does (Kitson and Constantinou 2024). Real-data evaluation leans on very few datasets, above all the Sachs protein-signalling data, whose reference graph is not definitive (Brouillard et al. 2025; Sachs et al. 2005). CausalArena collects synthetic, semantic and formula-grounded models with a public leaderboard (Li et al. 2026), and CausalBench scores gene-network inference on held-out perturbation data (Chevalley et al. 2025; Replogle et al. 2022). Our framework instead targets reusable contracts for independently auditing external systems, with benchmark suites as applications of those contracts.

2.2 Language models as a source of causal knowledge

Language models answer pairwise causal questions about named variables well, and part of that skill is memorised (Kiciman et al. 2023); they may repeat causal statements without performing causal inference (Zečević et al. 2023). When a model is treated as an imperfect expert, the guarantee that the true graph is kept holds only under an assumed error model, and in the authors' own case study the model was poorly calibrated and the true graph was excluded on one benchmark (Long et al. 2023). Asking for a causal order instead of individual edges avoids confusing direct with indirect effects (Vashishtha et al. 2025). Controls for memorisation now exist: neutral aliases in the model's queries (Kitson and Constantinou 2026), graphs published after a model's training cut-off (Srivastava et al. 2025), random graphs carrying real concept names (Li and Russo 2026), and names moved onto the wrong columns with the data unchanged, under which models' accuracy collapses (Dong et al. 2025). The last is the control we use; we add the layout controls and apply both to deployed agents and to benchmarks.

2.3 Causal agents and how they are evaluated

Causal-Copilot automates the whole analysis and states that the model's recommendations are soft suggestions that may not override the high-confidence decisions of its bootstrap procedure (Wang et al. 2025). ALCM, MATMCD and Causal Modelling Agents let a model refine or constrain a data-driven graph (Khatibi et al. 2024; Shen et al. 2025; Abdulaal et al. 2024), and open libraries expose model-based causal suggestions (CausalIQ 2026; PyWhy 2026). These systems are mostly evaluated on well-known named benchmarks against the plain algorithm. The only independent evaluation of Causal-Copilot we found scores it as a baseline and, when necessary, steered it towards a tractable algorithm because it often preferred intractable ones (Tagliapietra et al. 2026). Repeated runs of the same agent can disagree, so single-run comparisons can misrank systems (Mehta 2026). A position paper argues that agents should not supply edges or causal conclusions at all and asks for audit trails (Zheng et al. 2026). Our study combines recorded stage diagnostics with name and layout controls where supported; its independent auditing interface is intended to extend across externally developed systems.

Closest benchmark and auditing frameworks. Heuresis evaluates agentic causal discovery on anonymous simulated mechanisms using an interactive tool harness, algorithmic reference points and PC-centred trace decomposition; its appendices also report instrument-verification checks (Ma, Ding and Wang 2026). It supports multiple discovery algorithms, numerical probes and graph edits, so tracing and auditor checks alone do not distinguish our work (Ma, Ding and Wang 2026). Our target is an independent interface for externally developed systems, with declared information regimes, reference reliability and observation gaps across heterogeneous recorded audits; the scope demonstrated here remains narrower than that target. Causal-Audit uses time-series assumption diagnostics, calibrated risk and

uncertainty to guide method recommendation or abstention before discovery (Ruiz et al. 2026). This complements our focus on discovery-system behaviour under controlled inputs.

2.4 How much an edge can be trusted

Bootstrap resampling gives confidence measures for features of a learned network (Friedman, Goldszmidt and Wyner 1999), but the resulting edge frequencies can be miscalibrated (Kummerfeld and Rix 2019; Banerjee, Andrews and Kummerfeld 2025). Language models asked about direct edges give high stated confidence to most of their false-positive edges; agreement across prompts is better calibrated, though not significantly so (Kumar et al. 2026). An audit therefore has to test whether a system's confidence means anything, not assume it.

2.5 Auditing practice the protocol borrows

Metamorphic testing checks a program that has no oracle for each output by checking relations between the outputs of related inputs (Chen, Cheung and Yiu 1998; Segura et al. 2016); renaming or reordering columns while holding the numbers fixed is such a relation. Behavioural testing of language-processing models crosses capabilities with test types that include invariance tests, and finds failures that held-out accuracy hides (Ribeiro et al. 2020). Model cards and datasheets document a model's evaluation and a dataset's origin (Mitchell et al. 2019; Gebru et al. 2021), internal algorithmic audits define staged audit artefacts (Raji et al. 2020), and PROV-DM records which agent and activity produced a piece of data (Moreau and Missier 2013). Our certificate combines these ideas for causal claims: system identity, battery hash, every number with its source, and the command that reruns it.

3 Data

The audit draws on a suite built from CausalArena's semantic scenarios, Sachs and 4 other named datasets that ship with Causal-Copilot, and CausalBench's perturbation data. Table 1 gives their shapes, the reliability tier of each reference graph and the licences. All shapes are read from the suite manifests (Appendix A.2); the suites were frozen before any run and are checked by hash.

CausalArena semantic suite. CausalArena releases structural causal models whose variables carry names from a written scenario, with observational exports (Li et al. 2026). We take the 10 semantic scenarios of our test split, the first replicate of the observational-only release, 1,000 rows each. Each scenario is written 3 times with byte-identical numbers: under its real names, under neutral names, and under its own names moved onto the wrong

Table 1: Data used in the audit. Rows and variables of the files as given to the systems; for CausalBench, cells and perturbed genes after CausalBench's own filtering. Tier: reliability of the reference graph (Section 1).

Group	Dataset	Rows	Variables	Ref. edges	Tier	Licence
CausalArena	10 semantic scenarios	1,000	20–25	35–58	exact	CC BY 4.0
Named	Sachs	853	11	19	contested	none stated
	Abalone	4,177	8	7	provenance	CC BY 4.0
	Auto MPG	392	5	5	mismatched	CC BY 4.0
	Concrete strength	1,030	9	8	provenance	CC BY 4.0
	DWD climate	349	5	4	provenance	CC BY 4.0
CausalBench	K562	162,751	622	–	held-out interventions	CC BY 4.0
	RPE1	162,733	383	–	held-out interventions	CC BY 4.0

columns by a derangement chosen to reduce pairs adjacent under both readings (30 tables). Because the released column order is a causal order, the columns are shuffled with a fixed seed, and the file paths are opaque codes. The reference is the graph the data were simulated from, so scores are accuracy. The layout audit of the public leaderboard uses CausalArena's public files instead (250 semantic, 250 formula and 100 synthetic). The dataset is released under CC BY 4.0 on Hugging Face (`LAMDA-Tabular/CausalArena`).

Named datasets. Causal-Copilot ships Sachs and 4 other tables with reference graphs, and we use its copies (Wang et al. 2025). Sachs (Sachs et al. 2005) has 4 arms (the original, a copy under another path, neutral names, and moved names). Its reference is contested: the repository's graph has 19 edges, the bnlearn version 17, sharing 16 adjacencies with 1 reversed, and CausalArena uses a version with 20 edges, so a score on Sachs is agreement with one version, not accuracy. Abalone (Nash et al. 1994), Auto MPG (Quinlan 1993) and Concrete compressive strength (Yeh 1998) come from the UCI repository under CC BY 4.0, and the climate table matches the Deutscher Wetterdienst pairs of the Tübingen cause-effect benchmark (Mooij et al. 2016); DWD's open data are released under CC BY 4.0. Each has 3 arms (real, neutral, swapped). Their reference graphs are expert graphs with nothing to compare against (provenance tier), except Auto MPG, whose reference differs from CausalArena's version (4 variables, 3 edges, 398 rows), so its scores must not be set beside CausalArena's row. No licence is stated for the Sachs data.

CausalBench. CausalBench uses the CRISPRi Perturb-seq screens of Replogle et al. (2022) in the K562 and RPE1 cell lines and scores a predicted network on held-out interventions by a mean Wasserstein distance and a false omission rate (Chevalley et al. 2025). After CausalBench's own filtering, which we reproduced, the K562 data have 162,751 cells (10,691 unperturbed controls) and 622 perturbed genes, and RPE1 162,733 cells (11,485 controls) and 383 genes. The screens are released under CC BY 4.0 and the benchmark code under Apache 2.0. Our audit of this benchmark is cited from its result file, not re-run: all 4 published MeanDifference rows were reproduced; MeanDifference reaches 0.90–0.95

of the ceiling set by ranking pairs by the knockdown effects measured on the test split itself, so the main metric rewards ranking total effects; and its false omission rate stays within the pre-registered margin of random networks in 3 of the 4 cells, the remaining one being undefined.

Recorded runs. A recorded run keeps the system's identity (code hash checked before and after the run, model snapshot), every model request and reply, the graph after every stage, and the refinement's intermediate steps. Causal-Copilot was run unmodified (upstream commit `a31aa7c`) with gpt-4o-2024-08-06: 30 runs on the semantic suite in the name-arm study, and a main-control series of 30 runs on the same tables (API cost $12.57). The other systems' name-control results use a different semantic cohort and their own declared output/scoring contracts. Their results are not paired scenario-for-scenario with the Causal-Copilot collection.

Limits of the data. The semantic scenarios are simulated, so their exactness comes with simulation's artefacts; each table and arm was run once; one model family was used; and the named datasets' references are expert graphs. The protocol treats each of these as a declared regime or tier rather than hiding it.

Code

The reusable framework is published at `https://github.com/crucis-lab/crucis`, with documentation at `https://crucis-lab.github.io/crucis/`. Its implemented connection and experiment-definition contracts do not establish native readiness for every system or completion of the broader benchmark. The companion reproduction repository (`https://github.com/uzairgheewala/causal-copilot-q1`) recomputes the Causal-Copilot stage comparison from the recorded runs with one command and no API key, and documents the controls and metrics. The numbers in this report are read from result files of the research repository (`https://github.com/uzairgheewala/cc-research`) at commit `aef649241abe`; Appendix A.2 lists each with its file and JSON pointer.

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Appendices

A.1 Interpretation and implementation boundaries A1
A.2 Where every number comes from A2

A.1 Interpretation and implementation boundaries

The recorded name comparisons retain their own cohorts, stages and scoring rules. Their grouping under a shared decision rule is not a matched ranking of system architectures. Causal-Copilot's earlier collection is exploratory relative to that rule. Its paired completed-stage diagnostic asks how the same completed runs changed between stages; the displayed all-run initial mean and completed-run final mean have different denominators. A real-arm-only stage effect asks a further, different question and is not substituted for the paired aggregate.

The label-following comparison uses retained model-review changes in completed outputs, whereas the name-control score uses the graph handed to cycle removal, including observable incomplete runs. These endpoints are not interchangeable. The BND-08 rule's pre-run header names the pre-cycle endpoint, but its implemented calculation reads completed-output counts from the stability analysis. The preserved hash manifest binds the rule's bytes; it does not resolve that endpoint discrepancy or independently establish public pre-registration chronology. No endpoint or threshold has been changed here. The label comparison is conditional on an edge distinguishing the reference readings. It is a descriptive edge-level diagnostic with BND-08's recorded decision rule; edges from a shared scenario are not independent trials of general model behaviour. Its verdict therefore does not establish a population-wide causal effect of language-model review. The MATMCD finding likewise retains its point-rule interpretation even though the difference interval includes zero. The swapped-name construction seeks low reference overlap; optimality is not certified for every scenario.

The implemented connection and experiment-definition contracts preserve supplied evidence and make prerequisites checkable. Synthetic conformance tests assess those contracts. Recorded compatibility and numerical replay assess their historical integrations. Neither establishes fresh native execution, complete observation of arbitrary systems, or scientific validity merely from well-formed metadata. Execution and collection, validation of the auditor itself, and a new multi-system campaign remain distinct development obligations.

The companion reproduction covers the recorded Causal-Copilot comparison. The research repository retains the broader numerical source mappings, while the framework reposi-

tory owns reusable implementation. These scopes should be read together; the companion repository is not a reproduction of every claim in this report.

A.2 Where every number comes from

Generated by `docs/paper/q1/build_numbers.py` from `numbers.json`. Paths are relative to the root of cc-research; a pointer is a JSON pointer, where `*` takes every child and `[k=v]` the first list item with that field. A flag after a source marks a file that was not unchanged in git when the report was built.

Key	Value	Source file and JSON pointer (cc-research)
`cc.arms`	30	`analysis/results/fork/bld06_live_compare.json /arms`
`cc.completed`	22	`analysis/results/fork/bld06_live_compare.json /completed/main`
`cc.s0-adj`	0.584	`analysis/results/fork/bld06_live_compare.json /means/main/S0_adj_f1`
`cc.final-adj`	0.321	`analysis/results/fork/bld06_live_compare.json /means/main/final_adj_f1`
`cc.usd`	$12.57	`analysis/results/fork/bld06_live_compare.json /usd/main`
`cc.cyc.entered`	8	`analysis/results/fork/bld06_cycles.json /arms/main/runs_entering_cycle_step`
`cc.cyc.keyerr`	8	`analysis/results/fork/bld06_cycles.json /arms/main/cycle_step_keyerrors`
`bnd08.runs`	30	len of: `analysis/results/stability_s2/bnd08_summary.json /runs`
`bnd08.m3.labels`	81	`analysis/results/stability_s2/bnd08_summary.json /M3/labels_only`
`bnd08.m3.data`	20	`analysis/results/stability_s2/bnd08_summary.json /M3/data_only`
`bnd08.m3.n`	101	`analysis/results/stability_s2/bnd08_summary.json /M3/n`
`bnd08.model`	gpt-4o-2024-08-06	`analysis/results/stability_s2/bnd08_summary.json /snapshots/0`
`ab06.agents`	4	len of: `analysis/results/ab06/ab06.json /agents`
`ab06.cc-entered`	1	count_true of: `analysis/results/ab06/ab06.json /causal_copilot/*/counted_in_P9`
`ab06.systems`	5	add of: key `ab06.agents`; key `ab06.cc-entered`
`ab06.p9.entered`	4	len of: `analysis/results/ab06/ab06.json /P9/systems_entered`
`ab06.p9.losing`	3	len of: `analysis/results/ab06/ab06.json /P9/losing_ge_0.10`
`ab06.p9.min-scen`	5	`analysis/results/ab06/ab06.json /P9/min_paired_scenarios`

Key	Value	Source file and JSON pointer (cc-research)
ab06.p9.threshold	0.10	plans/ab06_prereg.json /predictions/P9/supported (pattern lose ≥ ([0-9.]+))
ab06.causaliq.drop	0.251	analysis/results/ab06/ab06.json /agents/causaliq/name_reliance_real_minus_swapped/mean
ab06.cc.drop	0.167	analysis/results/ab06/ab06.json /causal_copilot/causal_copilot_gpt4o/name_reliance_real_minus_swapped/mean
ab06.matmcd.drop	0.122	analysis/results/ab06/ab06.json /agents/matmcd/name_reliance_real_minus_swapped/mean
ab06.vashishtha.drop	0.020	analysis/results/ab06/ab06.json /agents/vashishtha/name_reliance_real_minus_swapped/mean
ab06.matmcd.dv1	0.6855	analysis/results/ab06/ab06.json /agents/matmcd/data_only_floor_real_arm/dv1_precision_final
ab06.matmcd.dv1-data	0.7199	analysis/results/ab06/ab06.json /agents/matmcd/data_only_floor_real_arm/dv1_precision_data_only
ab01.pc.published	0.333	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=PC (upstream, default rule);family=semantic]/published
ab01.pcmaxp.controlled	0.463	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=PC (upstream, maxP);family=semantic]/ours_shuffled_expectation
ab01.fci.authors	0.578	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=FCI;family=semantic]/ours_released_authors_rule
ab01.fci.expect	0.357	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=FCI;family=semantic]/ours_released_expectation
ab01.fci.gain	0.221	sub of: key ab01.fci.authors; key ab01.fci.expect
ab01.files.semantic	250	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=DirectLiNGAM;family=semantic]/files
ab01.files.formula	250	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=DirectLiNGAM;family=formula]/files
ab01.files.synthetic	100	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=DirectLiNGAM;family=synthetic]/files
ben13.reproduced	4	analysis/results/ben13/summary.json /reproduced_rows

Key	Value	Source file and JSON pointer (cc-research)
ben13.cells	4	count_notnull of: analysis/results/ben13/summary.json /lines/*/controls/*/placement_share_wasserstein
ben13.for.random-level	3	count_notnull of: analysis/results/ben13/summary.json /lines/*/controls/*/?placement_share_for_note
ben13.echo.min	0.90	min of: analysis/results/ben13/summary.json /predictions/P3/values/*
ben13.echo.max	0.95	max of: analysis/results/ben13/summary.json /predictions/P3/values/*
s2.families	10	len of: datasets/suite_q1s2/MAPPING.json /families
s2.tables	30	len of: datasets/suite_q1s2/MAPPING.json /arms
s2.arms-per	3	div of: key s2.tables; key s2.families
s2.rows	1,000	max of: datasets/suite_q1s2/MAPPING.json /families/*/rows
s2.vars.min	20	min of: datasets/suite_q1s2/MAPPING.json /families/*/variables
s2.vars.max	25	max of: datasets/suite_q1s2/MAPPING.json /families/*/variables
s2.edges.min	35	min of: datasets/suite_q1s2/MAPPING.json /families/*/edges
s2.edges.max	58	max of: datasets/suite_q1s2/MAPPING.json /families/*/edges
s1.sachs.rows	853	datasets/MAPPING.json /source/rows
s1.sachs.vars	11	len of: datasets/MAPPING.json /source/columns
s1.sachs.edges	19	len of: datasets/MAPPING.json /source/reference_edges
s1.sachs.arms	4	len of: datasets/MAPPING.json /arms
s1.sachs.bnlearn	17	analysis/results/reference_reliability/reliability.json /families/sachs/repo_vs_bnlearn/edges_b
s1.sachs.shared	16	analysis/results/reference_reliability/reliability.json /families/sachs/repo_vs_bnlearn/shared_adjacencies
s1.sachs.reversed	1	analysis/results/reference_reliability/reliability.json /families/sachs/repo_vs_bnlearn/reversed
s1.sachs.arena	20	analysis/results/reference_reliability/reliability.json /families/sachs/causalarena_version/edges
s1.abalone.rows	4,177	datasets/suite_b1/MAPPING.json /families/abalone/rows
s1.abalone.vars	8	len of: datasets/suite_b1/MAPPING.json /families/abalone/columns
s1.abalone.edges	7	len of: datasets/suite_b1/MAPPING.json /families/abalone/reference_edges

Key	Value	Source file and JSON pointer (cc-research)
s1.autompg.rows	392	datasets/suite_b1/MAPPING.json /families/autompg/rows
s1.autompg.vars	5	len of: datasets/suite_b1/MAPPING.json /families/autompg/columns
s1.autompg.edges	5	len of: datasets/suite_b1/MAPPING.json /families/autompg/reference_edges
s1.concrete.rows	1,030	datasets/suite_b1/MAPPING.json /families/concrete/rows
s1.concrete.vars	9	len of: datasets/suite_b1/MAPPING.json /families/concrete/columns
s1.concrete.edges	8	len of: datasets/suite_b1/MAPPING.json /families/concrete/reference_edges
s1.dwd.rows	349	datasets/suite_b1/MAPPING.json /families/dwd/rows
s1.dwd.vars	5	len of: datasets/suite_b1/MAPPING.json /families/dwd/columns
s1.dwd.edges	4	len of: datasets/suite_b1/MAPPING.json /families/dwd/reference_edges
s1.autompg.arena.vars	4	analysis/results/reference_reliability/reliability.json /families/autompg/causalarena_vars_edges_rows/0
s1.autompg.arena.edges	3	analysis/results/reference_reliability/reliability.json /families/autompg/causalarena_vars_edges_rows/1
s1.autompg.arena.rows	398	analysis/results/reference_reliability/reliability.json /families/autompg/causalarena_vars_edges_rows/2
s1.named.tables	12	len of: datasets/suite_b1/MAPPING.json /arms
s1.named	4	len of: datasets/suite_b1/MAPPING.json /families
s1.named.arms-per	3	div of: key s1.named.tables; key s1.named
cb.k562.cells	162,751	external/causalbench/dataset_k562_filtered.npz.json /cells_out *(excluded)*
cb.k562.genes	622	external/causalbench/dataset_k562_filtered.npz.json /genes_out *(excluded)*
cb.k562.controls	10,691	external/causalbench/dataset_k562_filtered.npz.json /controls *(excluded)*
cb.rpe1.cells	162,733	external/causalbench/dataset_rpe1_filtered.npz.json /cells_out *(excluded)*
cb.rpe1.genes	383	external/causalbench/dataset_rpe1_filtered.npz.json /genes_out *(excluded)*
cb.rpe1.controls	11,485	external/causalbench/dataset_rpe1_filtered.npz.json /controls *(excluded)*